DECENT COACHING CLASSES

X SB

Mind Your Math - IX

Time: 2 Hours **Maximum Marks: 40**

Note:

(1) All questions are compulsory.
(2) Use of calculator is not allowed. Figures drawn for the questions need not be to scale.
(3) The numbers on the right side indicate full marks.
(4) In case of MCQs, write the alphabet of the correct answer only.
(5) For Q.5, use of a ruler, compass and protractor is necessary. Keep construction lines visible.

Q.1 (A) and (B) – [8 marks]

Q.1 (A) Four alternative answers are given for each sub-question. Choose the correct one. **[4]**

(1) The areas of two similar triangles are 81 cm² and 49 cm². The ratio of their corresponding sides is
(a) 81 : 49 (b) 9 : 7 (c) 7 : 9 (d) 3 : 7

(2) Which of the following is a Pythagorean triplet?
(a) (5, 12, 14) (b) (8, 15, 17) (c) (7, 24, 26) (d) (9, 40, 42)

(3) The radius of a circle is 5 cm and a point P is 13 cm from the centre. The length of the tangent segment from P to the circle is
(a) 8 cm (b) 10 cm (c) 12 cm (d) 18 cm

(4) The slope of the line passing through the points (2, 3) and (4, 7) is
(a) 1/2 (b) 2 (c) 4 (d) −2

Q.1 (B) Solve the following sub-questions. **[4]**

(1) If $\sin\theta = 3/5$ and θ is an acute angle, find $\cos\theta$. [1]
(2) ABCD is a cyclic quadrilateral. If $\angle A = 75°$, find $\angle C$. [1]
(3) Find the distance between the points (−2, 3) and (4, −5). [1]
(4) Find the length of an arc of a circle of radius 12 cm whose central angle is 60°. ($\pi = 3.14$) [1]

Q.2 (A) and (B) – [8 marks]

Q.2 (A) Complete any TWO of the following activities. **[4]**

(1) In △ABC, seg DE ∥ side BC, with D on AB and E on AC. If AD = 3, DB = 5 and AE = 6, find EC. [2]

Solution: seg DE ∥ side BC ∴ AD/DB = _____ (Basic Proportionality Theorem)
∴ 3/5 = 6/EC ∴ EC = _____

(2) In △ABC, $\angle B = 90°$, $\angle C = 30°$ and AC = 14 cm. Find AB and BC. [2]

Solution: In a 30°–60°–90° triangle, side opposite 30° = ½ × _____
∴ AB = _____ cm and BC = $(\sqrt{3}/2) \times$ AC = _____ cm

(3) Find the coordinates of the point P which divides the segment joining A(1, 2) and B(7, 8) internally in the ratio 2 : 1. [2]

Solution: $x = (m \cdot x_2 + n \cdot x_1)/(m + n) = (2 \times ____ + 1 \times ____)/3 = ____$
$y = (m \cdot y_2 + n \cdot y_1)/(m + n) = (2 \times ____ + 1 \times ____)/3 = ____$ ∴ P(____ , ____)

Q.2 (B) Solve any TWO of the following. **[4]**

(1) The areas of two similar triangles are 25 cm² and 36 cm². If a side of the smaller triangle is 10 cm, find the corresponding side of the larger triangle. [2]

(2) Chords AB and CD of a circle intersect at a point P inside the circle. If AP = 6, PB = 4 and CP = 3, find PD. [2]

(3) Find the slope of the line passing through the points A(−3, 4) and B(5, −2). [2]

(4) Prove that $\cos^2\theta\,(1 + \tan^2\theta) = 1$. [2]

Q.3 (A) and (B) – [12 marks]

Q.3 (A) Complete any ONE of the following activities. **[3]**

(1) From an external point P, tangent segments PA and PB touch a circle with centre O at points A and B. If ∠APB = 70°, find ∠AOB. [3]

Solution: Radius is perpendicular to the tangent at the point of contact.

∴ ∠OAP = _____° and ∠OBP = _____°

In quadrilateral OAPB: ∠AOB = 360° − (∠OAP + ∠OBP + ∠APB) = 360° − (____ + ____ + 70°) = _____°

(2) The radius of a circle is 7 cm and the central angle of a sector is 90°. Find the length of the arc and the area of the sector. ($\pi = 22/7$) [3]

Solution: Length of arc = $(\theta/360) \times 2\pi r$ = (____/360) × 2 × (22/7) × 7 = _____ cm

Area of sector = $(\theta/360) \times \pi r^2$ = (____/360) × (22/7) × ____ = _____ cm^2

Q.3 (B) Solve any THREE of the following. **[9]**

(1) In △ABC, AB = 11, AC = 7 and BC = 14. Find the length of the median AD. [3]

(2) Prove that the lengths of the two tangent segments drawn to a circle from an external point are equal. [3]

(3) Show that the points A(1, −1), B(5, 2) and C(9, 5) are collinear. [3]

(4) Prove that $1/(1 + \sin\theta) + 1/(1 - \sin\theta) = 2\sec^2\theta$. [3]

Q.4 – [8 marks]

Q.4 Solve any TWO of the following. **[8]**

(1) In △ABC, point D lies on AB and point E lies on AC such that seg DE ∥ side BC. If AD : DB = 3 : 4, find the ratio of the area of △ADE to the area of trapezium DBCE. [4]

(2) From the top of a building 24 m high, the angle of depression of the foot of a tower is 45° and the angle of elevation of the top of the tower is 60°. Find the height of the tower. ($\sqrt{3} = 1.73$) [4]

(3) A solid cone of radius 6 cm and height 24 cm is melted and recast into a solid sphere. Find the radius of the sphere. [4]

Q.5 – [4 marks]

Q.5 Do any ONE of the following constructions. **[4]**

(1) Construct △ABC in which BC = 6 cm, ∠B = 60° and AB = 5 cm. Then construct △A′BC′ similar to △ABC, whose sides are 7/5 of the corresponding sides of △ABC. [4]

(2) Draw a circle of radius 3 cm with centre O. Take a point P at a distance of 7 cm from O and construct the two tangents to the circle from P. [4]

— End of Question Paper —